

Hotel Management Booking System - UML Diagrams

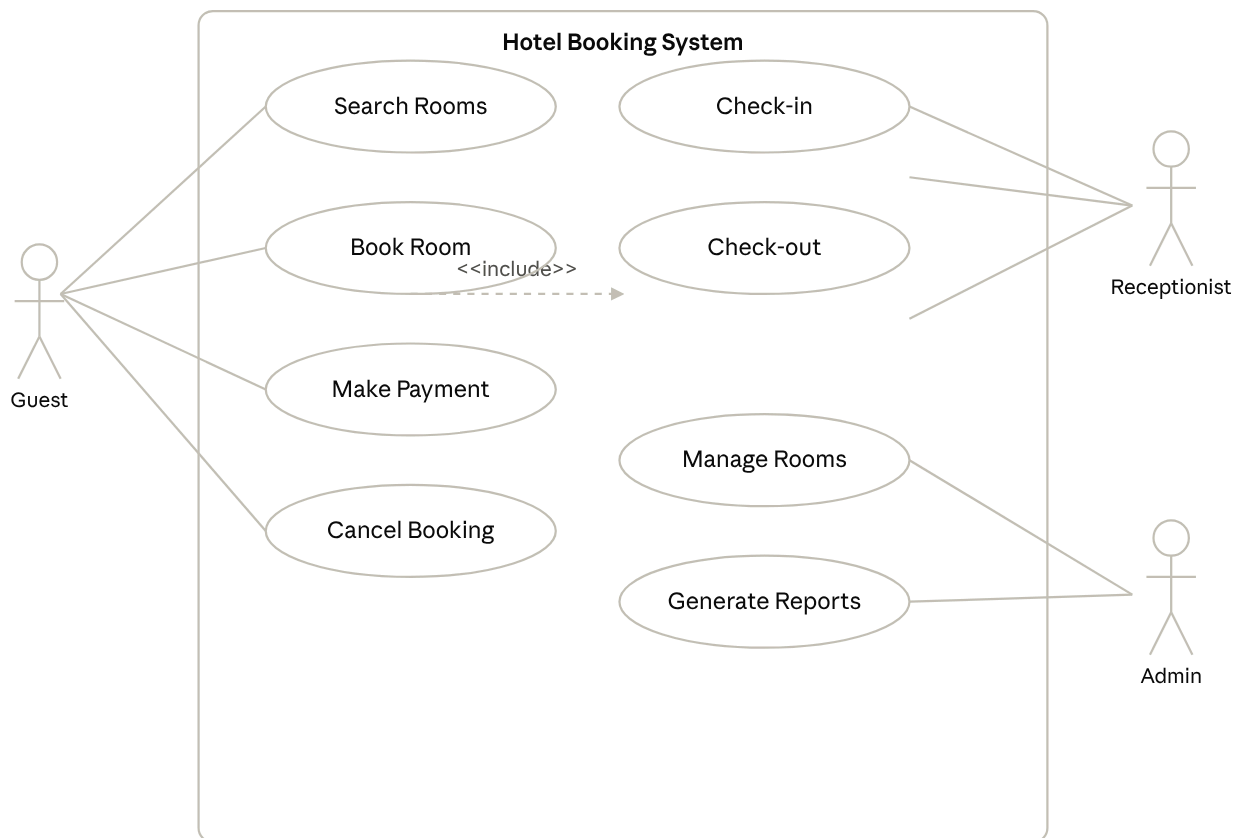
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Overview

This document presents the core UML diagrams for a **Hotel Management Booking System**: the actors and their interactions (use case), the object structure (class), the message flow for a booking (sequence and collaboration), the booking process (activity), and the lifecycle of a booking (state chart).

1. Use Case Diagram

Guests, receptionists and admins use eight functions of the system

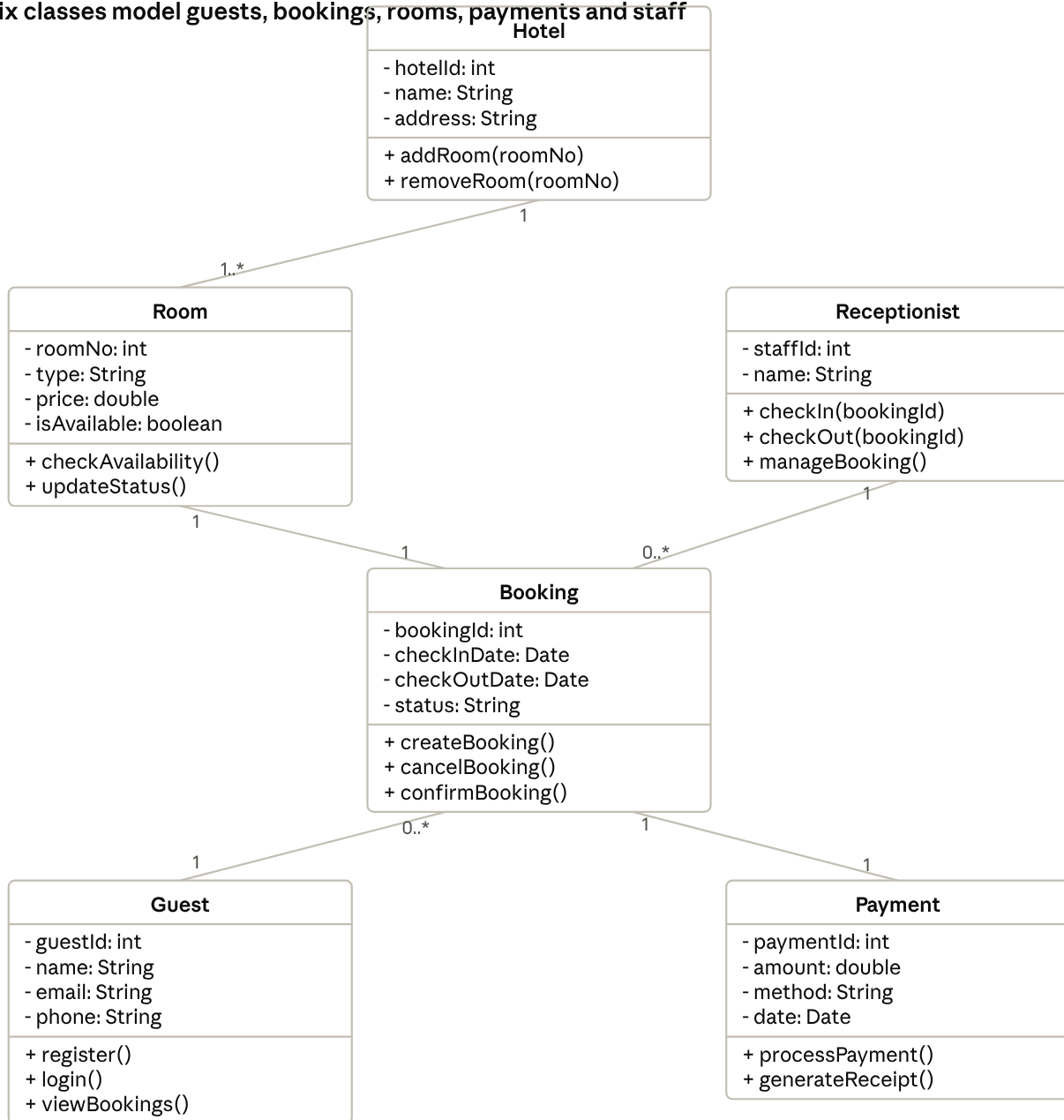


use case diagram · 3 actors, 8 use cases

The **Guest** searches rooms, books a room, makes a payment and can cancel a booking. Booking a room always includes making a payment (`<<include>>`). The **Receptionist** handles check-in, check-out and can also book a room on a guest's behalf. The **Admin** manages room inventory and generates reports.

2. Class Diagram

Six classes model guests, bookings, rooms, payments and staff

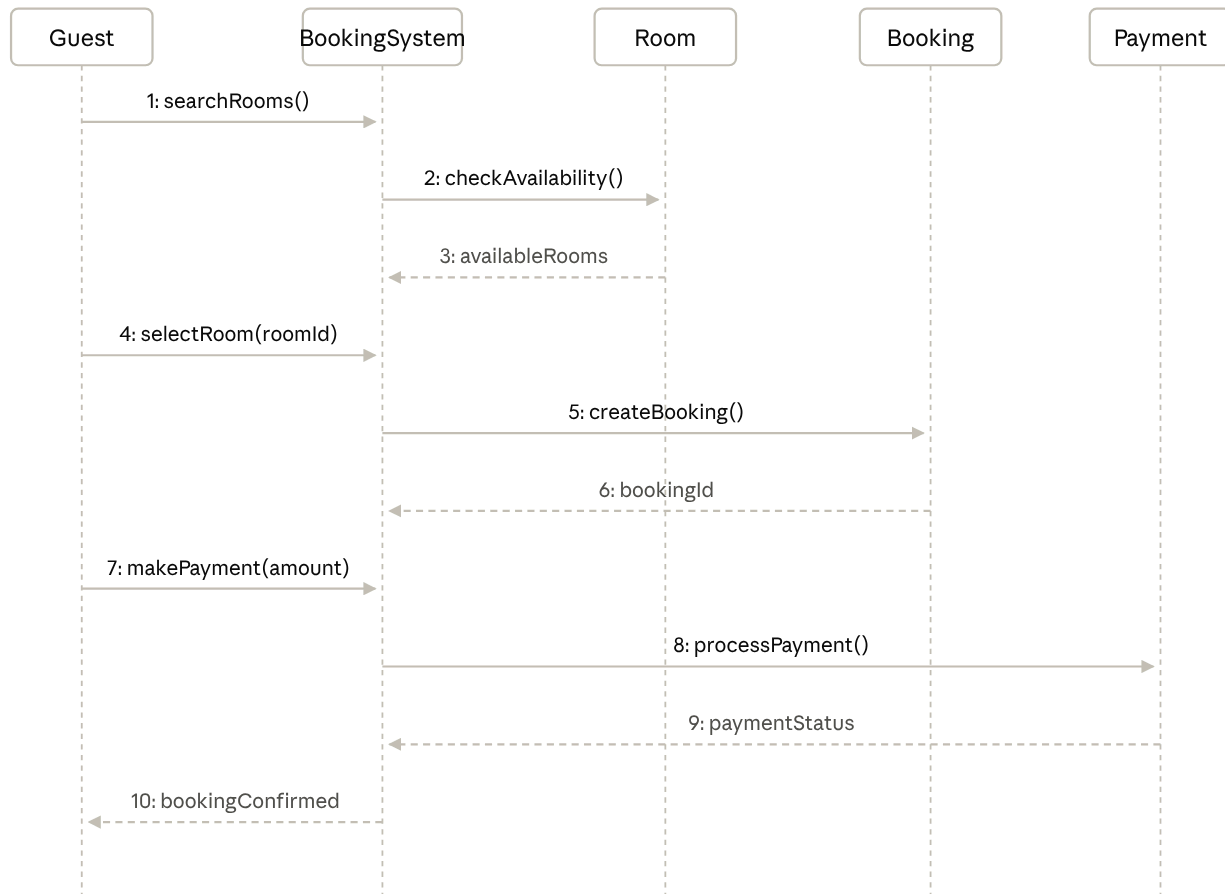


class diagram · 6 classes with attributes, methods and multiplicities

A **Hotel** contains one-to-many **Rooms**. Each **Room** is used in one **Booking**, and a **Receptionist** manages zero-or-more bookings. A **Guest** makes zero-or-more bookings, and each **Booking** has exactly one **Payment**.

3. Sequence Diagram

Ten messages flow from guest search to booking confirmation

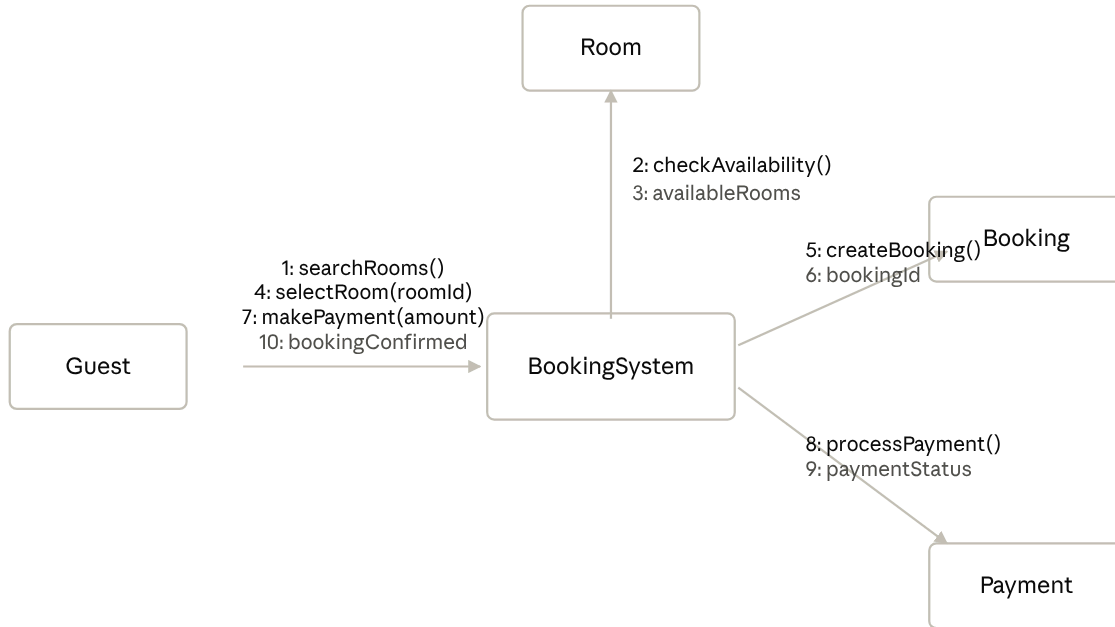


sequence diagram · 5 lifelines, 10 messages for the booking flow

The diagram traces one booking end to end: the Guest searches and selects a room through the BookingSystem, which checks the Room's availability, creates a Booking, and routes payment through the Payment object before confirming back to the Guest.

4. Collaboration (Communication) Diagram

Five objects exchange ten numbered messages around the booking system

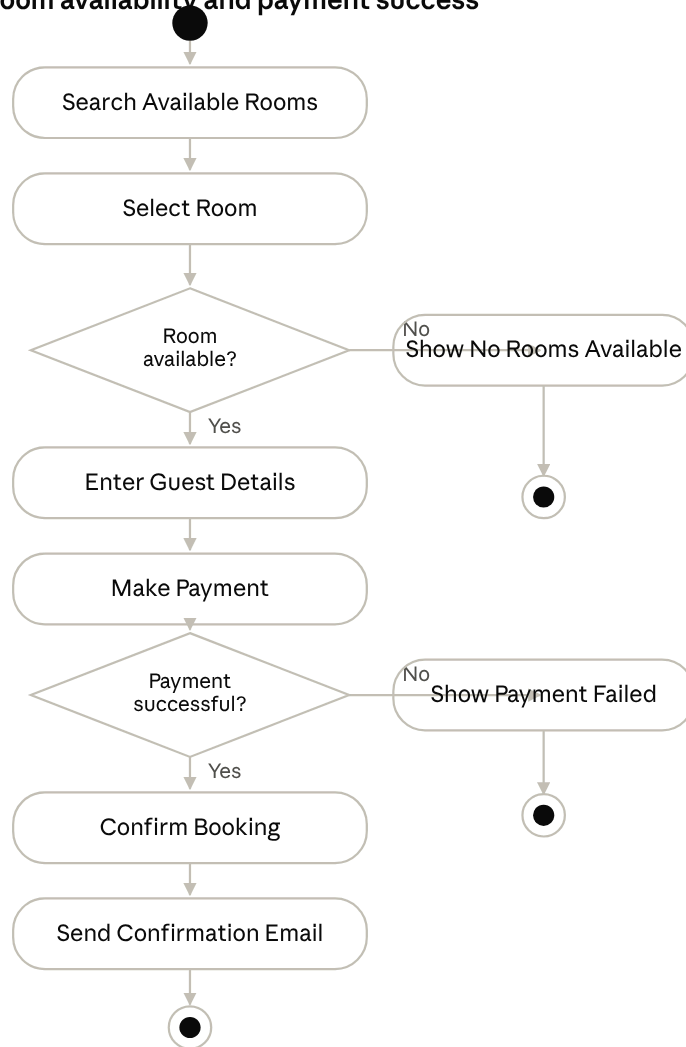


collaboration diagram · 5 objects, numbered messages around the booking system

The same ten messages as the sequence diagram are shown here as numbered links between objects, emphasising how the BookingSystem sits at the centre coordinating the Guest, Room, Booking and Payment objects rather than the order in which messages occur.

5. Activity Diagram

The booking activity branches on room availability and payment success

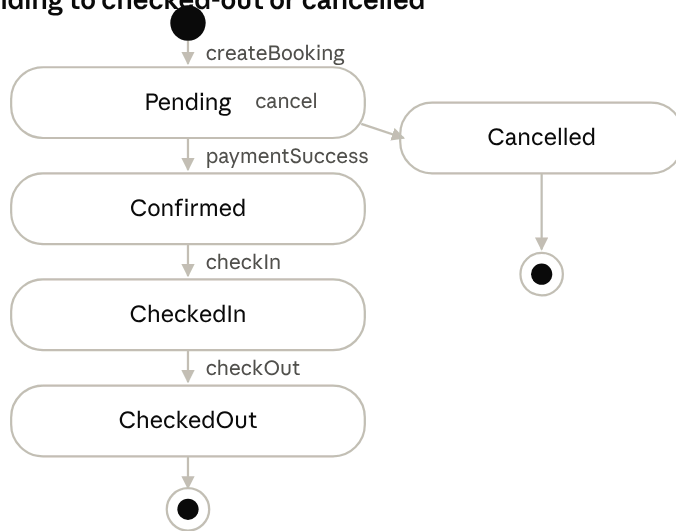


activity diagram · start, 7 actions, 2 decisions, 3 end nodes

The booking activity starts with a room search and ends one of three ways: no rooms available, payment failure, or a confirmed booking with an emailed confirmation.

6. State Chart (State Machine) Diagram

Six states carry a booking from pending to checked-out or cancelled



state chart · 6 states tracking one booking's lifecycle

A booking starts **Pending** once created. It moves to **Confirmed** on successful payment, or to **Cancelled** if the guest cancels before paying. A confirmed booking proceeds through **CheckedIn** and **CheckedOut**, which is a final state; **Cancelled** is also a final state.